

Week 2 Day 2

Testing Techniques

1) Black Box Testing

The testing done without looking at the internal code, only care about input and output. You treat the system as "black-box" where you don't know what is inside (QA/Test Eng).

For eg: Testing a vending machine you put in money and press a button and you check if the right snack comes out. You don't care or know about internals.

i) Equivalence Partitioning

- Group similar inputs together and test just one from each group since they should all behave the same way.

For eg: An app accept age from 1-100 for registration. Instead of testing each and every number, You test one valid age (25), one invalid low value (-5) and one invalid high value like (150).

ii) Boundary Value Analysis

- It test the boundary values specifically like for eg: There is a discount applies for order above 500 then you check 499, 500, 501

↑ ↑ ↑
 invalid valid valid

iii) Decision Table Testing

It's a testing technique used when the output depends on multiple conditions.

for eg: The loan approval system if (Income > 50k AND Credit Score > 700) → Approve, else → Reject.

Rule	Card Valid	Pin Correct	Enough Balance	Action
1	Yes	Yes	Yes	get cash
2	Yes	Yes	No	show not Enough Balance
3	Yes	No	Yes	show incorrect Pin
4	Yes	No	No	show incorrect Pin
5	No	-	-	show invalid Card.

iv) State Transition Testing

It shows how a system behaves as it moves between different states when an event/action happens.

for eg: - Valid Transition

ATM (Idle) → Card Inserted → PIN Entered → Select cash amount → Cash Dispensed.

Invalid Transition

ATM (Idle) → Card Inserted → Incorrect PIN Entered (3 times) → Card blocked

2) White Box Testing

The testing technique where the tester looks at the internal code, logic, structure and ~~flow~~ flow of the program.

For eg: A mechanic who knows the internal wiring, each connection and all the electrical paths not just whether the cars start or not.

i) Statement Testing

It make sure that every executable statement in the code run or executed at least once. For eg: If a function has 10 line : run test until all 10 have been triggered.

ii) Branch Testing.

It make sure every decision path is tested, both true and false.

For eg: if (age $\geq$ 18) allow entry, else deny
You test both the 'true' branch and the 'false' branch.

3) Experience Based Techniques

The testing done through your own experience, intuition and knowledge from past ~~mistake~~ mistake.

For eg- A doc with 20 years of experience in there field can tell what's wrong with patient with just few tests.

i) Error Guessing

With the past experience the tester to guess what is most likely causing the bug.

for eg: The experience personal knows when make a email field you know user often paste extra spaces into the field at the end or start.

ii) Exploratory Testing.

In this testing we learn about the app while testing it, with no fixed script.

for eg: Tester open new shopping app and just click around, add items remove them, trying weird combos to see what breaks.

iii) Checklist Based Testing

This testing use checklist with is build from past experience.

for eg: A pro tester knows what are the function that cause most problems so the tester build his checklist based on that.

① EP BVA Example for phone number

Equivalence partitioning ~~help~~ help us to ~~red~~ reduce the number of test need to do like we do not have to go through every single phone number you can make separate group for each ~~ity~~ type of number provided.

Partition	Eg	Expected
Less than 10	974284622	Invalid
Exactly 10	9742846227	valid
More than 10	9742862276	invalid
Non-numeric	9AB7282276	invalid
Empty	" "	invalid

BVA since the phone number must be exactly 10 we need to find Boundary value for 10

BVA	Numbers	Expectation
less than 10	9	Invalid
Exactly 10	10	valid
More than 10	11	Invalid

State transition Diagram

